

Pankaj Maulekhi

AI/ML Engineer — Generative AI & Agentic Systems

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🔗 pankaj-2708 🌐 pankaj-maulekhi 🌐 My Portfolio

Professional Summary

AI/Data Science undergraduate at IIITDM Kurnool specializing in generative modeling and agentic AI systems. Built and published PM-MiniFinLLM, a 110M-parameter finance LLM pretrained from scratch on 2.36B tokens with SFT and DPO alignment. Core skills include LLM training and fine-tuning, and image generation models (diffusion, GANs). Experienced in MLOps and cloud deployment using AWS (SageMaker, ECR, EC2), Docker, and CI/CD pipelines for production-grade generative AI systems.

Technical Skills

- **Languages:** Python, C++, C, SQL
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Plotly, Seaborn, BeautifulSoup
- **Machine Learning:** Scikit-Learn, XGBoost, LightGBM, Feature Engineering, Hyperparameter Tuning
- **Deep Learning:** PyTorch, TensorFlow (Keras), Transformers, Computer Vision, NLP, Distributed Training (DDP)
- **Generative AI:** Diffusion Models, Flow Matching, GANs, Hugging Face, Fine-Tuning, Quantization
- **Agentic AI and RAG:** LangChain, LangGraph, LangSmith, MCP, RAG, Vector Databases, Agentic Pipelines, Ragas
- **MLOps and Deployment:** MLflow, DVC, Optuna, FastAPI, Docker, GitHub Actions (CI/CD), AWS (S3, EC2, ECR, SageMaker, Bedrock)

Education

- **B.Tech in Artificial Intelligence and Data Science**, IIITDM Kurnool 2024 – Present
CGPA: 8.18 (4th Semester)
- **Class XII**, Jawahar Navodaya Vidyalaya Almora (CBSE) 2024
Percentage: 95.4%
- **Class X**, Jawahar Navodaya Vidyalaya Almora (CBSE) 2022
Percentage: 96.4%

Achievements

- Achieved **98.38 percentile** in JEE Mains 2024, competing among over 1.4 million candidates nationwide.
- Solved **400+ problems** on LeetCode, covering Data Structures, Algorithms, and Dynamic Programming.

Projects

- **PM-MiniFinLLM — Finance LLM from Scratch** 🔗 GitHub
Tech: PyTorch, HuggingFace Transformers, TRL, DPO, PyTorch DDP, FP16 Mixed Precision
 - Architected a 110M-parameter GPT-2 style decoder-only transformer with RoPE positional embeddings, covering the full LLM pipeline from pre-training to alignment
 - Pre-trained on 2.36B tokens from SEC EDGAR filings using Distributed Data Parallel (DDP) training across 2× NVIDIA T4 GPUs with FP16 mixed precision
 - Fine-tuned on 28K financial QA samples via HuggingFace SFTTrainer and aligned using a custom DPO training loop; publicly registered on HuggingFace Hub
- **Fine-Tuning FLUX.2-klein-9B with DreamBooth & 4-bit QLoRA on AWS SageMaker** 🔗 GitHub
Tech: PyTorch, HuggingFace Diffusers/PEFT, bitsandbytes (4-bit QLoRA), AWS SageMaker, Flow Matching
 - Customized Black Forest Labs' 9B-parameter FLUX.2-klein diffusion transformer via DreamBooth and 4-bit NF4 QLoRA, capping peak GPU memory at 20.8% (~5GB/24GB) on a single AWS SageMaker `ml.g5.xlarge` A10G instance
 - Implemented a from-scratch Flow Matching training loop with custom latent patchification and 3D space-time positional coordinate construction, plus offline multi-layer text-encoder hidden-state pre-encoding that saved ~6GB VRAM and boosted throughput by ~45%
 - Orchestrated a 4-phase cloud MLOps pipeline (prompt pre-encoding, QLoRA training, benchmark generation, comparative inference) on SageMaker with stateful checkpoint resuming, completing 50 epochs (650 steps) in ~49 minutes
- **Self-RAG & Corrective RAG for the Indian Constitution & IPC** 🔗 GitHub
Tech: LangGraph, LangChain, DeepSeek R1 & V3, AWS Bedrock, ChromaDB, Sentence-Transformers, DeepEval, MLflow, DVC, Docker
 - Designed a 16-node LangGraph state machine implementing Self-RAG and Corrective RAG (CRAG) for hallucination-resistant QA over the Constitution of India and the IPC, with DeepSeek R1 chain-of-thought query routing across retrieval, web search, and direct generation
 - Engineered parallel multi-query vector retrieval with fan-out/fan-in via LangGraph's Send API and Map-Reduce relevance filtering, backed by ChromaDB and Amazon Titan Embed Text v2, with automatic Tavily web search fallback and rolling-summary conversational memory
 - Developed a three-tier DeepEval evaluation framework (component, dataset generation, pipeline) tracked via MLflow on DagsHub and reproduced with DVC, achieving 0.96 contextual precision, 0.95 faithfulness, 0.94 answer relevancy, and 0.93 contextual recall; deployed via Docker Compose on AWS EC2 with a GitHub Actions CI/CD pipeline to ECR